Create Project Folder

```
albraa@albraa-Cyborg-15-A13UDX:~$ mkdir docker-compose-project
albraa@albraa-Cyborg-15-A13UDX:~$ cd docker-compose-project
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$ mkdir nginx data
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$ touch docker-compose.ym
```

Create docker-compose.yml

```yaml
version: "3.9"

services:
  nginx:
    image: nginx:latest
    container_name: nginx-server
    ports:
      - "8080:80"
    volumes:
      - ./nginx/default.conf:/etc/nginx/conf.d/default.conf:ro
    networks:
      - app-network
    depends_on:
      - mysql

  mysql:
    image: mysql:8.0
    container_name: mysql-db
    restart: always
    environment:
```

```
    container_name: mysql-db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpassword
      MYSQL_DATABASE: myappdb
      MYSQL_USER: myuser
      MYSQL_PASSWORD: mypassword
    volumes:
      - mysql-data:/var/lib/mysql
    networks:
      - app-network

networks:
  app-network:
    driver: bridge

volumes:
  mysql-data:
    driver: local
```

: Create NGINX Config

```
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$ touch nginx/default.con
f
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$ nano nginx/default.conf
```

Start Docker Compose

```
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$ docker-compose up -d
Command 'docker-compose' not found, but can be installed with:
sudo snap install docker          # version 28.4.0, or
sudo apt  install docker-compose  # version 1.29.2-6
See 'snap info docker' for additional versions.
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$ sudo snap install docke
r
[sudo] password for albraa:
docker 28.4.0 from Canonical✓ installed
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$ docker-compose up -d
WARN[0000] /home/albraa/docker-compose-project/docker-compose.yml: the attribute
 `version` is obsolete, it will be ignored, please remove it to avoid potential
confusion
[+] Running 20/20
 ✔ mysql Pulled                                                          333.1s
   ✔ 74a8e4bbd9fe Pull complete                                          118.9s
   ✔ 8814198b348b Pull complete                                          119.0s
   ✔ 0644bff8d158 Pull complete                                          119.0s
   ✔ bfb2a1b21556 Pull complete                                          119.2s
   ✔ 7aee539d1d90 Pull complete                                          119.2s
   ✔ 3ca1b3148b2f Pull complete                                          119.2s
   ✔ d6b535e661c2 Pull complete                                          183.7s
   ✔ 25e72585209f Pull complete                                          183.7s
```

Verify MySQL Volume

```
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS
      PORTS                                       NAMES
b2360156f1c2   nginx:latest   "/docker-entrypoint.…"   4 minutes ago   Up 4 minu
tes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   nginx-server
4c6932e42353   mysql:8.0      "docker-entrypoint.s…"   4 minutes ago   Up 4 minu
tes   3306/tcp, 33060/tcp                       mysql-db
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$ docker volume ls
DRIVER    VOLUME NAME
local     docker-compose-project_mysql-data
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$ echo "<h1>Hello from NG
INX + MySQL!</h1>" > index.html
docker cp index.html nginx-server:/usr/share/nginx/html/index.html
bash: !: event not found
lstat /home/albraa/docker-compose-project/index.html: no such file or directory
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$ # Option 1: Escape the
!
echo "<h1>Hello from NGINX + MySQL\!</h1>" > index.html

# Option 2: Remove the !
echo "<h1>Hello from NGINX + MySQL</h1>" > index.html
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$ echo "<h1>Hello from NG
INX + MySQL</h1>" > index.html
```

```
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$ docker cp index.html ng
inx-server:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to nginx-server:/usr/share/nginx/html/index.html
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$ nano docker-compose.yml

albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-project$
```

localhost:8080

Hello from NGINX + MySQL

2-

Create Project Folder

```
albraa@albraa-Cyborg-15-A13UDX:~$ mkdir docker-compose-python-redis
albraa@albraa-Cyborg-15-A13UDX:~$ cd docker-compose-python-redis
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-python-redis$ nano app.py
```

Python App

```
  GNU nano 7.2                          app.py *
import time
import redis
from flask import Flask

app = Flask(__name__)

cache = redis.Redis(host='redis', port=6379)

def get_hit_count():
    retries = 5
    while True:
        try:
            return cache.incr('hits')
        except redis.exceptions.ConnectionError as exc:
            if retries == 0:
                raise exc
            retries -= 1
            time.sleep(0.5)

@app.route('/')

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

Create requirements.txt

```
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-python-redis$ nano requirements.
txt
```

```
  GNU nano 7.2                     requirements.txt
flask
redis
```

Create Dockerfile

```
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-python-redis$ nano Dockerfile
```

```
  GNU nano 7.2                          Dockerfile *
# Use official Python image
FROM python:3.11-slim

# Set working directory
WORKDIR /app

# Copy requirements and install
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Copy app code
COPY app.py .

# Expose port 5000
EXPOSE 5000

# Run the app
CMD ["python", "app.py"]
```

Create Docker Compose File

```
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-python-redis$ nano docker-compos
e.yml
```

```
  GNU nano 7.2                    docker-compose.yml *
version: "3.9"

services:
  web:
    build: .
    container_name: python-flask-app
    ports:
      - "5000:5000"
    depends_on:
      - redis
    networks:
      - app-network

  redis:
    image: redis:7.0
    container_name: redis-server
    ports:
      - "6379:6379"
    networks:
      - app-network
```

Build and Start Containers

```
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-python-redis$ docker-compose up
-d --build
WARN[0000] /home/albraa/docker-compose-python-redis/docker-compose.yml: the attr
ibute `version` is obsolete, it will be ignored, please remove it to avoid poten
tial confusion
[+] Running 9/9
 ✔ redis Pulled                                                          38.8s
   ✔ a2318d6c47ec Pull complete                                          35.6s
   ✔ b59068c7715f Pull complete                                          35.6s
   ✔ 96796a2514ff Pull complete                                          35.6s
   ✔ b8c430d0189b Pull complete                                          35.7s
   ✔ fcf1b4feeffd Pull complete                                          35.8s
   ✔ bcec16c47e35 Pull complete                                          35.8s
   ✔ 4f4fb700ef54 Pull complete                                          35.8s
   ✔ 73586727588c Pull complete                                          35.9s
#1 [internal] load local bake definitions
#1 reading from stdin 561B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 338B done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/python:3.11-slim
```

Test the App

```
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-python-redis$ curl http://localh
ost:5000
Hello World! I have been seen 1 times.
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-python-redis$
```

Stop and Remove Containers

```
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-python-redis$ docker-compose dow
n
WARN[0000] /home/albraa/docker-compose-python-redis/docker-compose.yml: the attr
ibute `version` is obsolete, it will be ignored, please remove it to avoid poten
tial confusion
[+] Running 3/3
 ✔ Container python-flask-app                      Removed                10.3s
 ✔ Container redis-server                          Removed                 0.3s
 ✔ Network docker-compose-python-redis_app-network Removed                 0.2s
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-python-redis$
```

```
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-python-redis$ docker-compose dow
n -v
WARN[0000] /home/albraa/docker-compose-python-redis/docker-compose.yml: the attr
ibute `version` is obsolete, it will be ignored, please remove it to avoid poten
tial confusion
[+] Running 1/1
 ✔ Volume docker-compose-python-redis_redis-data  Removed                  0.0s
albraa@albraa-Cyborg-15-A13UDX:~/docker-compose-python-redis$
```